Reward Shaping and Curriculum PPO for Sparse-Reward SoccerTwos Control

Vedaang Chopra
Georgia Institute of Technology
United States
vedaangchopra@gatech.edu

Abstract: This project studies PPO-based policy learning in SoccerTwos, a sparse-reward Unity soccer environment where agents must learn to reach the ball, move it toward goal, and score with limited feedback. A plain PPO baseline learned slowly, so I compared it against two modifications: reward shaping with small dense bonuses for ball-directed progress, and curriculum learning with progressively harder start states. Reward shaping improved early training from 0.2399 to 0.4036 mean episode reward at the same 2M-step budget, but curriculum training produced the strongest policy. The final curriculum v3 agent, exported as `TEAMNAME_v3_AGENT`, reached 1.9692 final mean episode reward after about 30M environment steps and won 8 of 10 recorded matches against the PPO baseline. These results suggest that the main bottleneck was sparse-reward exploration, and that curriculum learning was the most effective way to address it.

Keywords: Reinforcement Learning, SoccerTwos, Curriculum Learning

1 Overview

SoccerTwos is a two-team Unity soccer environment where cube agents must learn to move, approach the ball, defend, and score goals through interaction. The task is challenging because the default reward is sparse: many trajectories contain little useful feedback unless the agent reaches meaningful soccer events such as scoring or conceding. I treated this as a policy-learning problem and used PPO as the base algorithm because it is stable, widely used for continuous interaction environments, and supported by the RLlib training pipeline used in the project. The main goal was to test whether additional structure could make PPO learn better soccer behavior. I therefore compared a sparse-reward PPO baseline, a reward-shaped PPO agent with dense progress signals, and a curriculum PPO agent that starts from easier states before training on harder game situations. The final submitted policy, `TEAMNAME_v3_AGENT`, was produced by extending the best curriculum PPO checkpoint to a longer training budget.

2 Method Description

PPO baseline. I used Proximal Policy Optimization (PPO) as the baseline policy-gradient method. The baseline agent was trained with the default sparse SoccerTwos reward and a feed-forward ReLU actor-critic network. PPO was a reasonable starting point because it constrains policy updates through a clipped objective, which helps prevent large unstable updates from rollout data. In the baseline and reward-shaped runs, the policy/value model used one hidden layer of width 512 with shared value layers.

Reward-shaped PPO. The first modification was to add dense reward shaping during training. The SoccerTwos task is difficult because useful rewards are sparse: an agent may move for many steps without receiving clear feedback about whether it is getting closer to useful soccer behavior. To reduce this exploration burden, I wrapped the training environment with a shaping term computed

Table 1: Main PPO training configurations used in the project.

Method	Reward	Curriculum	Network
PPO baseline	sparse	no	[512]
Reward-shaped PPO	sparse + shaping	no	[512]
Curriculum PPO	sparse + curriculum setup	yes	[256, 256]
Curriculum PPO v3	resumed curriculum	yes	[256, 256]

from the environment `info` dictionary. Let d_{pb} denote the player-to-ball distance and d_{bg} denote the ball-to-goal distance. At each transition, the wrapper adds

$$r_{\text{shape}} = \text{clip}\left(0.01(d_{pb}^{t-1} - d_{pb}^{t}) + 0.02(d_{bg}^{t-1} - d_{bg}^{t}), -0.05, 0.05\right).$$

This gives a small positive bonus when the controlled player moves closer to the ball or when the ball moves closer to the opponent goal. The clipping keeps the shaping reward small relative to actual scoring outcomes, so the shaped signal guides exploration without replacing the main task objective. This shaping is used only during training; the exported policy does not require privileged `info` fields at inference time. This design follows the general motivation of reward shaping for sparse-reward RL [1].

Curriculum PPO. The second modification was curriculum learning. Instead of training immediately from the full difficult game distribution, the curriculum starts the agent in easier situations and gradually increases the difficulty. A callback sets the initial ball and player states through the Unity environment channel and advances through five tasks, from "Very Easy Goal" to "Random Players," when the running `episode_reward_mean` exceeds 1.5. This makes early exploration more productive: the agent first learns simple ball-to-goal behavior, then transfers that behavior to harder starting states. The curriculum policy also used a deeper ReLU MLP with hidden layers $[256, 256]$ and shared value layers. The final v3 policy resumed the strongest curriculum checkpoint and extended training to about 30.0M environment steps, following the curriculum-learning idea that easier early tasks can improve learning in sparse-reward settings [2].

Final hyperparameters. The submitted policy was trained with Ray RLlib PPO using the PyTorch backend. The final v3 run used a learning rate of 5×10^{-5}, discount factor $\gamma = 0.99$, PPO clipping parameter 0.3, 30 SGD epochs per update, train batch size 80,000, minibatch size 8,000, rollout fragment length 2,000, gradient clipping at 0.5, 40 rollout workers, and 1 GPU. The exported submission package was `TEAMNAME_v3_AGENT`.

3 Preliminary Results

Figure 1 summarizes the main training result. Plain PPO struggled under the sparse SoccerTwos reward and reached only 0.2399 final mean episode reward after 2.0M environment steps. Adding reward shaping improved the same-budget result to 0.4036, which suggests that dense progress feedback helped exploration but was not sufficient by itself. The largest improvement came from curriculum learning: the first curriculum run reached 1.7869 mean reward after 2.06M steps, while the extended curriculum v3 run reached 1.9692 final mean reward, with a best logged mean reward of 1.9705 at 29.85M steps. Overall, the curves indicate that curriculum design was the most important factor for learning useful SoccerTwos behavior.

Table 2 reports the recorded headless evaluation matches for the final curriculum v3 policy. Because the number of evaluation episodes is small, these results should be treated as preliminary rather than definitive. Still, the v3 policy won 8 of 10 matches against the PPO baseline and 6 of 10 matches against the earlier curriculum policy, which is consistent with the learning-curve improvement.

Figure 3 shows the detailed training diagnostics for the final curriculum v3 run. The mean episode reward quickly approached 2.0 and remained stable, while episode length dropped sharply and then stabilized. This suggests that the final policy learned to reach terminal soccer outcomes more efficiently. The PPO diagnostics also show decreasing entropy and value loss, which is consistent with the policy becoming less random and the critic fitting the return signal more accurately.

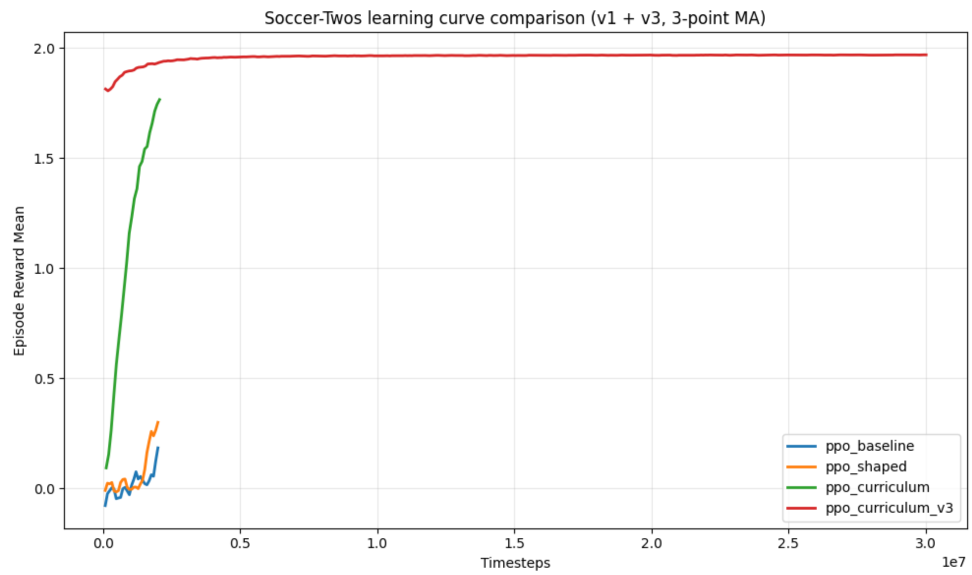

Figure 1: Mean episode reward over training for baseline PPO, reward-shaped PPO, curriculum PPO, and the extended curriculum v3 policy. Reward shaping improves over the sparse-reward baseline, but curriculum training gives the largest gain.

Table 2: Recorded headless evaluation summaries for the final curriculum v3 policy. Match counts are small, so these results are preliminary.

Matchup	Episodes	Wins	Agent mean	Opponent mean
Curriculum v3 vs. baseline	10	8	1.1140	-1.2084
Curriculum v3 vs. curriculum	10	6	0.3407	-0.4258

4 Reward, Observation, and Architecture Modifications

Reward modification. The main reward-level change is the training-only `RewardShapingWrapper`. Instead of relying only on sparse scoring rewards, the wrapper uses `player_info` and `ball_info` to compute progress signals: whether the player moves closer to the ball and whether the ball moves closer to the opponent goal. The shaping bonus is clipped to ± 0.05 and added to the environment reward during training. This modification improved the 2.0M-step mean episode reward from 0.2399 for baseline PPO to 0.4036 for reward-shaped PPO, suggesting that dense progress feedback helped early exploration, although it was not sufficient to produce the strongest final policy.

Observation and action handling. I did not implement a custom observation-space wrapper or Unity-side observation rewrite in the final policy. The agent uses the provided 336-dimensional SoccerTwos observation vector during training. The main interface handling is action-related: the training pipeline uses a flattened `Discrete(27)` action space, while live SoccerTwos matches use the branched `MultiDiscrete([3,3,3])` action format. Because no custom observation features were added, this report does not claim an observation modification.

Architecture and curriculum modification. The final policy differs from the baseline in both network structure and training schedule. The baseline and reward-shaped PPO agents use a single hidden layer of width 512, while the curriculum PPO agents use a two-layer $[256, 256]$ ReLU MLP with shared value layers. More importantly, curriculum training changes the distribution of initial states: the agent first trains on easier near-goal situations and then progresses to harder ball and player placements. This curriculum was the most effective modification in the project, producing substantially higher learning rewards and better recorded match outcomes than baseline PPO.

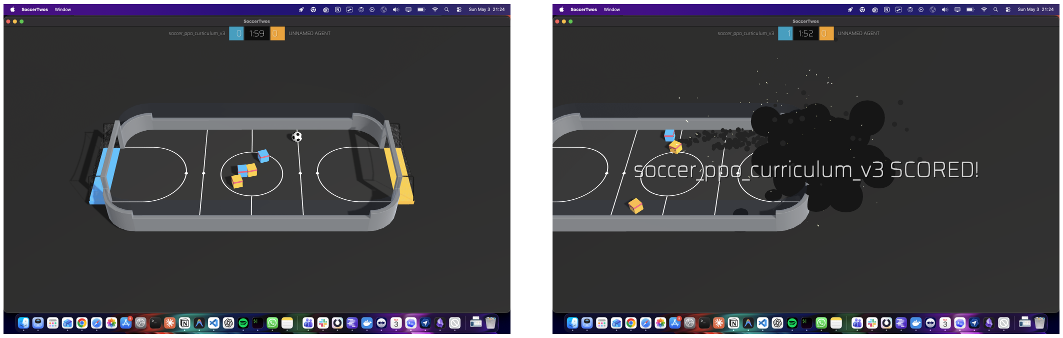

Figure 2: Gameplay screenshots from the final `TEAMNAME_v3_AGENT` evaluated in the SoccerTwos environment. The agent (blue team) consistently moves toward the ball and positions itself to score.

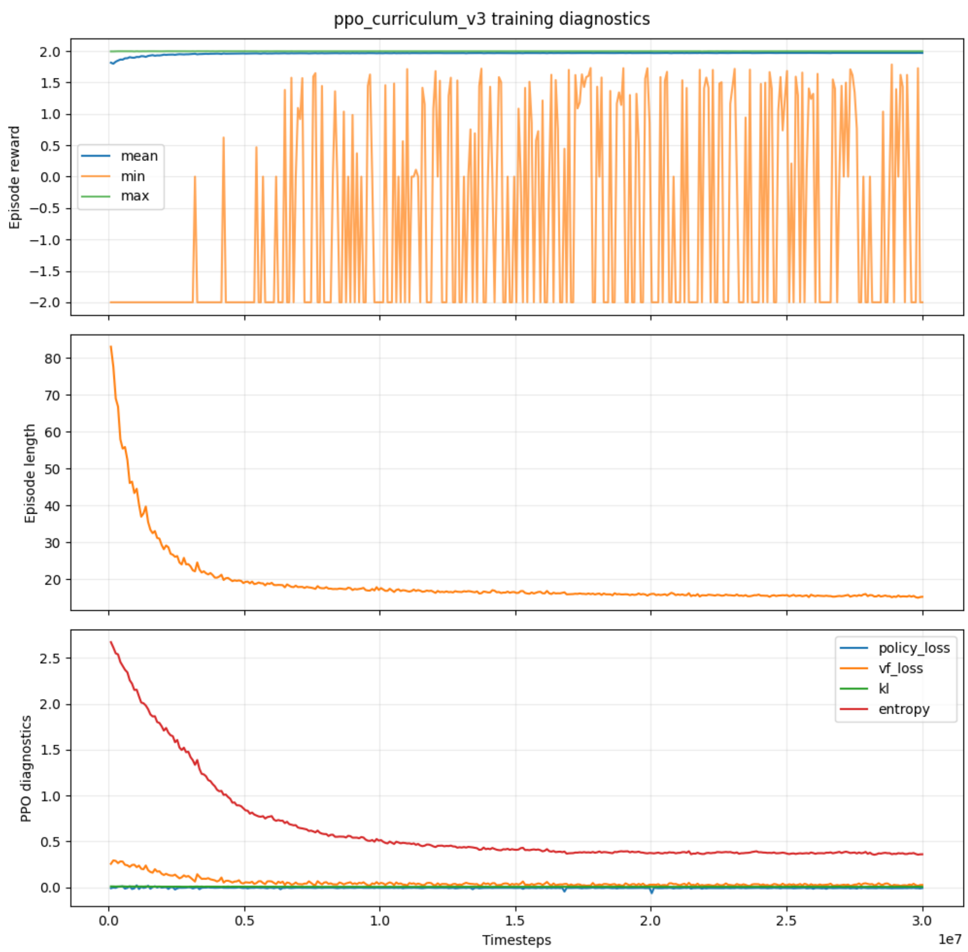

Figure 3: Detailed diagnostics for the final `ppo_curriculum_v3` run. Mean reward stabilizes near 2.0, episode length decreases substantially, and PPO diagnostics show stable convergence behavior.

5 Limitations and Future Work

The main limitation is evaluation size. The strongest recorded comparison against the PPO baseline contains 10 headless episodes, not a larger 100-episode evaluation, so the match results should be interpreted as preliminary. Training was also performed in a simplified single-player setting against a fixed opponent policy, so the final agent should not be interpreted as having learned full multi-agent coordination such as passing or team positioning. Imitation learning was explored as an additional direction, but it was not used as the final successful method. Future work should run larger evaluations against random, baseline, and TA agents; compare reward-shaped PPO directly against curriculum v3 in head-to-head matches; and test whether the curriculum policy remains robust under self-play or stronger multi-agent opponents.

6 GitHub Link

https://github.com/Vedaang-Chopra/RL_Soccer_project

References

[1] A. Y. Ng, D. Harada, and S. Russell. Policy invariance under reward transformations: Theory and application to reward shaping. In *Proceedings of the Sixteenth International Conference on Machine Learning*, pages 278–287, 1999.

[2] Y. Bengio, J. Louradour, R. Collobert, and J. Weston. Curriculum learning. In *Proceedings of the 26th Annual International Conference on Machine Learning*, pages 41–48, 2009.